



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to P S K AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

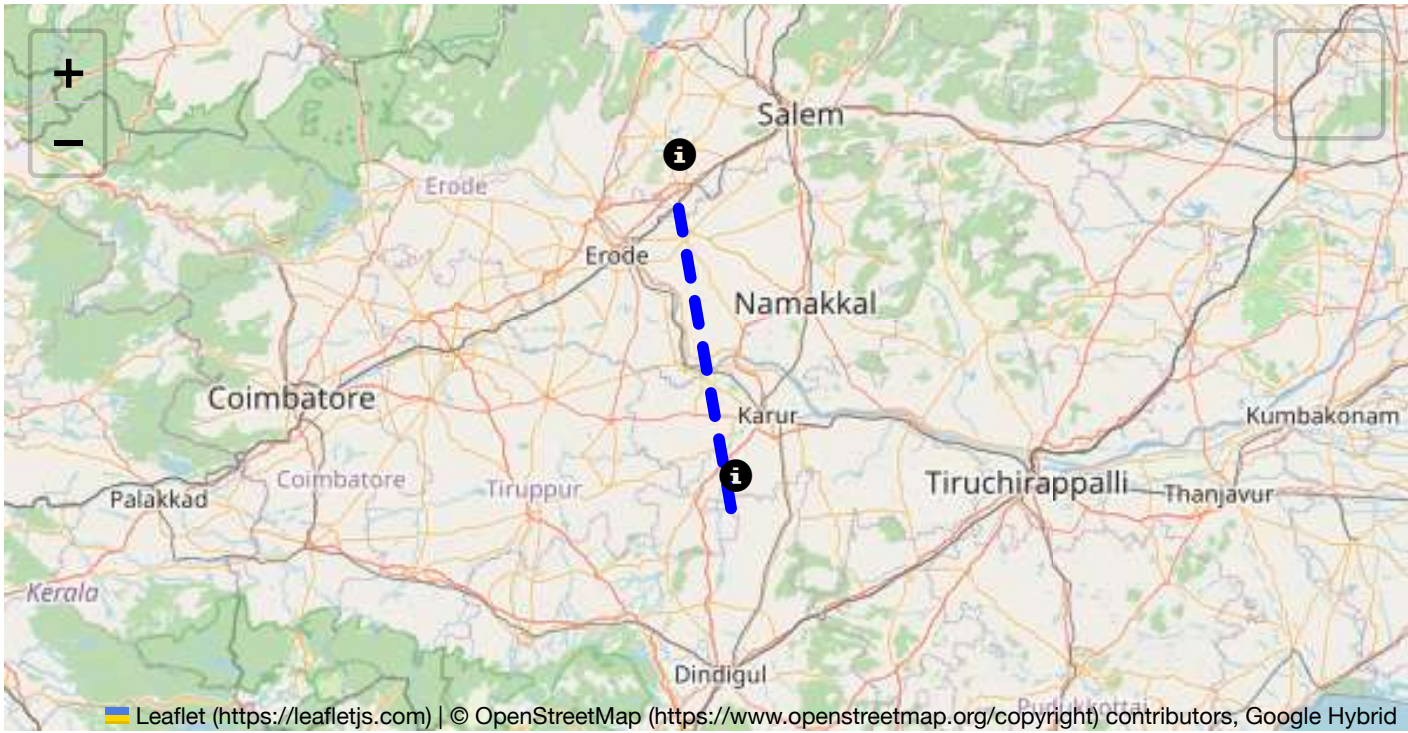
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

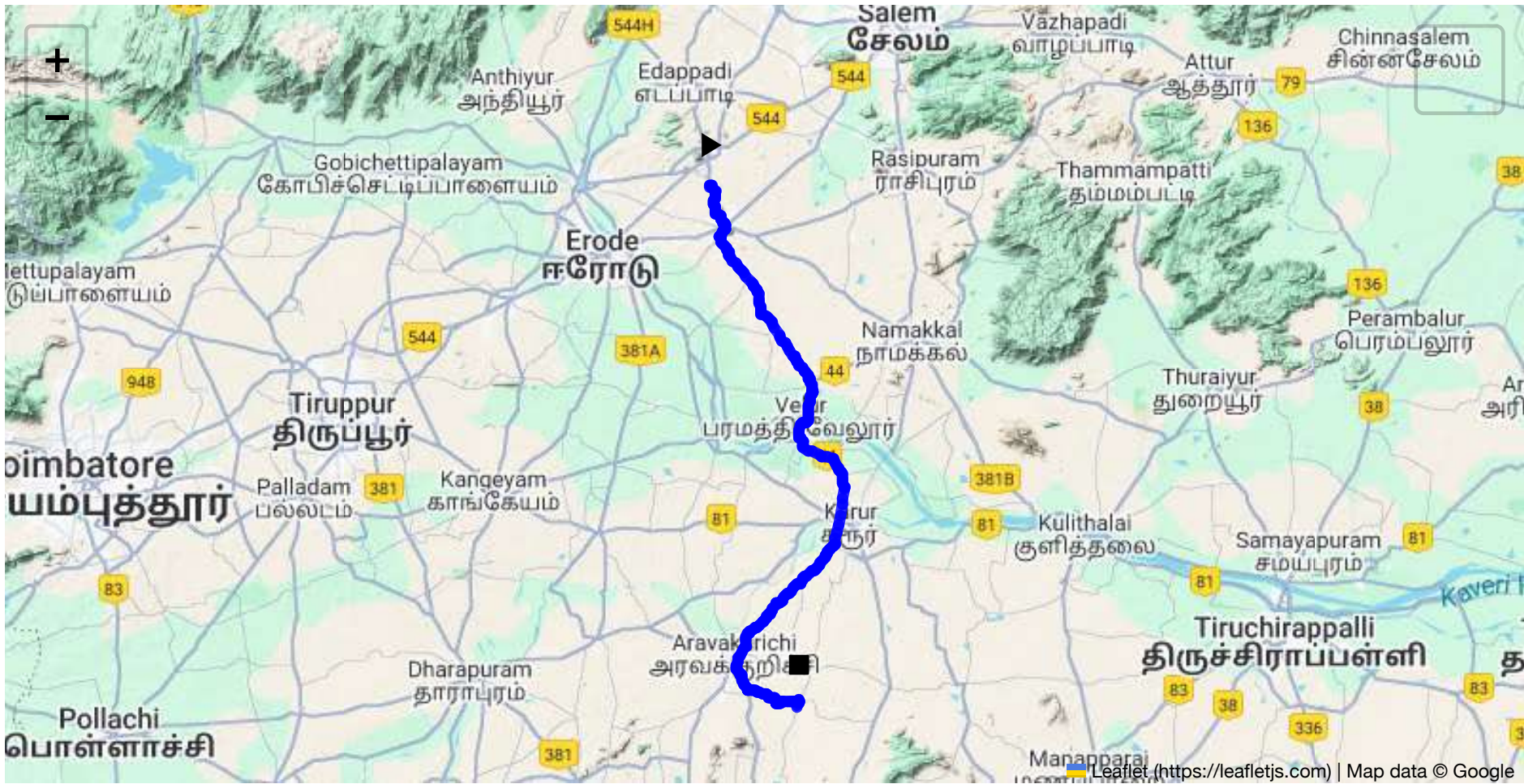
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 108.52 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.5 hours

Start: (11.4381, 77.8734)
End: (10.699354, 78.000307)



Welcome to the Journey Risk Management Study

Here's a comprehensive analysis of the route from Sangagiri, Tamil Nadu to Esanatham, Tamil Nadu, specifically aimed at preparing a truck driver for transporting hazardous materials.

- 1. Overview of the Route Map:
 - The direct route spans approximately 108.52 kilometers and connects two rural areas in Tamil Nadu. It involves traveling through a mix of state highways and local roads. The primary state highways used may include SH-20 and SH-175, among others.
- 2. Typical Weather Conditions and Potential Weather-Related Hazards:
 - Tamil Nadu typically experiences tropical weather, with the monsoon season between October and December bringing heavy rainfall. The likelihood of road flooding and reduced visibility during this period is high. During peak summer (April to June), heat can lead to road surface damage and tire blowouts.
- 3. Analysis of Traffic Patterns:
 - Peak hours often coincide with local business and school timings, typically from 8:00 AM to 10:00 AM and 5:00 PM to 7:00 PM. Expect increased congestion near major towns and intersections.
 - Market areas, especially on weekends, can also cause slowed traffic.
- 4. Assessment of Road Quality and Infrastructure:
 - Generally, state highways are well-maintained, but local roads can be uneven with potential potholes. Watch for narrow bridges and lack of shoulders in rural sections, especially in agricultural zones.
- 5. Suggestions for Alternative Routes for Emergencies:
 - If the primary route is blocked or unsafe, consider using NH-44, which could be a longer detour but offers better infrastructure and services.
 - Mapping apps and GPS should be updated for real-time traffic data and alternative paths.

6. Summary of Local Regulations Affecting Hazardous Material Transport:
- Drivers must adhere to specific transport regulations such as speed limits (60 km/h for heavy vehicles) and restricted travel through certain populated areas. Ensure all necessary permits are obtained prior to departure.
7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:
- Limited historical data on major incidents, but minor accidents due to road conditions or driver error are not uncommon. Awareness and caution are advised, especially in less populated sections.
8. Environmental Considerations and Sensitive Areas:
- The route passes close to agricultural fields and small village communities. Efforts should be made to avoid spills and maintain noise control to minimize impact on local residents and the environment.
9. Analysis of Communication Coverage:
- Mobile network coverage is generally reliable along the state highways but may be patchy in remote rural areas. It is wise to carry multiple communication devices as a backup.
10. Estimated Emergency Response Times for Different Route Segments:
- In urban areas, emergency response can average 20 to 30 minutes but may exceed an hour in remote locations. It is crucial to have contingency plans and emergency kits available.
11. Overall Summary of Risk Assessment:
- The route presents moderate risks predominantly due to rural road conditions, potential weather impacts, and communication limitations. Adherence to local regulations and preparedness for emergencies can mitigate most risks effectively. It is vital to remain vigilant and adaptable to changing conditions, particularly concerning weather and traffic.

In conclusion, thorough planning, informed navigation, and awareness of regional hazards and regulations will support a safer journey. Drivers should regularly check local updates and ensure the vehicle is well-prepared for all anticipated scenarios.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	High	11.37862, 77.89488	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16950, 78.01544	30 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	Medium	11.10102, 78.00572	30 KM/Hr
11	Turn	Medium	11.05261, 78.05138	30 KM/Hr
12	Turn	Medium	10.89913, 78.03032	30 KM/Hr
13	Turn	Medium	10.79857, 77.92714	30 KM/Hr
14	Turn	Medium	10.78576, 77.92670	30 KM/Hr
15	Turn	High	10.72341, 77.92805	15 KM/Hr
16	Turn	High	10.72296, 77.93309	15 KM/Hr
17	Turn	Medium	10.72242, 77.93367	30 KM/Hr
18	Turn	Medium	10.71598, 77.95672	30 KM/Hr
19	Turn	High	10.71153, 77.96780	15 KM/Hr
20	Turn	Medium	10.70434, 77.96938	30 KM/Hr
21	Turn	High	10.70405, 77.96965	15 KM/Hr
22	Turn	Medium	10.70341, 77.97475	30 KM/Hr
23	Turn	Medium	10.70567, 77.97849	30 KM/Hr
24	Turn	Medium	10.70653, 77.97903	30 KM/Hr
25	Turn	Medium	10.70781, 77.98181	30 KM/Hr
26	Turn	Medium	10.70644, 77.99157	30 KM/Hr
27	Turn	Medium	10.70742, 77.99465	30 KM/Hr
28	Turn	Medium	10.70637, 77.99669	30 KM/Hr
29	Turn	Medium	10.70797, 78.00188	30 KM/Hr
30	Turn	High	10.70766, 78.00275	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16950, 78.01544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05261, 78.05138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.89913, 78.03032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79857, 77.92714



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.78576, 77.92670



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72341, 77.92805



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.72296, 77.93309



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.72242, 77.93367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.71598, 77.95672



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.71153, 77.96780



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70434, 77.96938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.70405, 77.96965



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70341, 77.97475



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70567, 77.97849



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70653, 77.97903



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70781, 77.98181



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70644, 77.99157



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70742, 77.99465



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70637, 77.99669



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.70797, 78.00188



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.70766, 78.00275

